

Group Policies – part 3

IT Infrastructure Management (W04IST-SI0826G)

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This week

After this class, you should be able to:

- configure software deployment via Group Policy using Assigned and Published methods
- configure folder redirection for user profile folders in GPMC
- create a Password Settings Object and explain how it overrides the domain password policy

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Why this matters

- A new application must be installed on 200 domain computers before Monday
- Visiting each machine manually takes days and produces inconsistent results
- A single GPO deploys silently at next startup – no user interaction, no admin visits
- Today: three GPO features every AD admin uses in production

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Agenda

- 1 Software Installation
- 2 Folder Redirection
- 3 Password Settings Objects
- 4 Wrapping Up

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Quick recall

Without looking at your notes:

- 1 Where is GPO configuration data stored in the domain? (two locations)
 - 2 What is the default order of GPO processing?
 - 3 Which tool shows what settings a specific computer actually received?
 - 4 What is UNC (Universal Naming Convention) or *network* path format?
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- 1 GPC in Active Directory + GPT in SYSVOL
 - 2 Local → Site → Domain → OU (last applied wins)
 - 3 **Group Policy Results** (gpresult)
 - 4 \\server\share\path\file.msi (not C:\path\file.msi)

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Group Policy software installation

- Dedicated Group Policy Client Side Extension
- Deploys software to **computers** or **users**
- Requires no administrative privileges from the end user
- One of the few policy areas that **cannot** be refreshed with gpupdate
 - Computer assignments: applied only at **startup**
 - User assignments: applied only at **logon**

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Supported package format

- Only .msi (Windows Installer) packages are supported natively
- .exe installers can be wrapped in a script, but require user permissions
- Modern alternative: **MSIX** — supported from Windows 10 / Server 2019

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Installation methods

Assigned (Computer or User)

- Installed automatically — no user choice
- Computer: installs at startup; User: installs at logon
- Cannot be removed by the user

Published (User only)

- Appears in **Control Panel > Programs** for manual install
- Also installs automatically when a user opens an associated file type
- User can uninstall the application

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Predict — Assigned or Published?



- Your organisation needs **all** domain computers to have a VPN client
- Users must not be able to uninstall it
- Which deployment method should you use — and on which node?
- **Assigned**, under **Computer Configuration**
- Installs at next startup, silently, for every computer in scope
- Published is not available under Computer Configuration

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Preparation for deployment



- Create a shared folder on a server accessible to all domain computers
 - Example: \\dc1\install\
 - Share permissions: **Domain Computers** — Read
- Copy the .msi file to the share
- Use only the UNC path in the GPO — **never** a local path

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Creating a software installation policy



- 1 GPMC → GPO → Edit
- 2 Computer: **Computer Configuration** > **Policies** > **Software Settings** > **Software installation**
- 3 User: **User Configuration** > **Policies** > **Software Settings** > **Software installation**
- 4 Right-click → **New** > **Package** → enter UNC path to .msi
- 5 Choose **Assigned** or **Published** in the deployment dialog

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Limitations of GPO software installation



- Active Directory does not track consumed licenses
- No built-in monitoring of installation status or failure
- Administrator is responsible for license compliance
- Suitable for small/medium environments

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Modern deployment alternatives



- **Microsoft Intune** — cloud-based MDM, replaces GPO deployment in hybrid/Azure AD environments
- **Microsoft Endpoint Configuration Manager (MECM/SCCM)** — enterprise-scale deployment with inventory and compliance reporting
- **MSIX** — modern packaging format; supports clean install/uninstall, auto-update

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Why redirect folders?



- By default, user profile folders (Documents, Desktop, Pictures...) are stored **locally**
- Problem: data is lost if the computer fails; inaccessible from other machines
- **Folder Redirection** transparently moves selected folders to a network share
- User experience unchanged — paths stay the same from the user's perspective

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How folder redirection works



- Configured per-folder under **User Configuration > Policies > Windows Settings > Folder Redirection**
- Two targeting options:
 - **Basic** — all users redirected to the same root path (e.g., \\srv01\home\%username%\Documents)
 - **Advanced** — different paths per security group membership
- GPMC creates the subfolder automatically if granted permission

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Folder redirection — practical considerations

- The network share must be accessible at logon — no redirection without connectivity
- **Offline Files (CSC)** can cache redirected folders locally for disconnected use
 - Enable via **Computer Configuration > Policies > Windows Settings > Scripts** or Offline Files policy
- Set NTFS permissions: user has **Full Control** of their own subfolder only
- Do not redirect **AppData(Roaming)** — this interacts with roaming profiles and causes conflicts

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Password policy — the problem

- Every GPO can contain password settings (minimum length, complexity, lockout...)
- But **only** password settings in a GPO linked **at domain level** take effect on accounts
- This means the entire domain shares one password policy — unsuitable for mixed environments
- Example:
 - 1 *service accounts* need longer passwords
 - 2 *contractors* need shorter lockout windows

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Fine-Grained Password Policy

- **Password Settings Objects (PSOs)** — introduced in Windows Server 2008
- Stored as AD objects in CN=Password Settings Container,CN=System,DC=...
- Applied directly to **users** or **security groups** — not to OUs
- Multiple PSOs can exist
- PSO with **lower** msDS-PasswordSettingsPrecedence value wins

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Creating a PSO – ADAC

- Active Directory Administrative Center > domain > System > Password Settings Container
- Right-click → New > Password Settings
- Configure: minimum length, complexity, lockout, history, precedence
- Add target users or groups under **Directly Applies To**

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Creating a PSO – PowerShell

```
1 New-ADFineGrainedPasswordPolicy `
2   -Name "ServiceAccounts-PSO" `
3   -Precedence 10 `
4   -MinPasswordLength 20 `
5   -ComplexityEnabled $true `
6   -LockoutThreshold 5 `
7   -LockoutDuration "00:30:00"
8 Add-ADFineGrainedPasswordPolicySubject `
9   -Identity "ServiceAccounts-PSO" `
10  -Subjects "SVC_Accounts"
11 Get-ADUserResultantPasswordPolicy -Identity SVC_Accounts
```

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Discuss – PSO design (2 min)

- You have three groups of users:
 - regular employees
 - service accounts
 - IT admins
- Regular: standard domain policy is fine
- Service accounts: 20-char password, no expiry
- IT admins: 16-char password, shorter lockout duration

- 1 How many PSOs do you need?
- 2 What precedence values would you assign?

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In the lab



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You will:

- Deploy an .msi package via GPO using Assigned deployment (Computer Configuration)
- Configure Documents folder redirection for a test user to a network share
- Create a Password Settings Object for the chosen group and verify it applies

Before you go



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On a piece of paper or in your notes, write:

- 1 One thing from today you are confident about
- 2 One thing you are still unsure about — bring it to the lab

References



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- [1] A. Warren, *Exam ref 70-742 identity with windows server 2016*. Microsoft Press, 2017.
- [2] O. Thomas, *Windows server inside out: Updated for windows server 2025*. Microsoft Press, 2025.
- [3] J. Krause, *Mastering windows server 2025 - fifth edition*. Packt Publishing, 2025.